
SF Salaries Exercise

Welcome to a quick exercise for you to practice your pandas skills! We will be using the [SF Salaries Dataset](#) from Kaggle! Just follow along and complete the tasks outlined in bold below. The tasks will get harder and harder as you go along.

**** Import pandas as pd.****

```
import pandas as pd
```

**** Read Salaries.csv as a dataframe called sal.****

```
sal = pd.read_csv('SF_Salaries.csv')
```

**** Check the head of the DataFrame. ****

```
   Id  EmployeeName
JobTitle \
0  1  NATHANIEL FORD  GENERAL MANAGER-METROPOLITAN TRANSIT
AUTHORITY
1  2  GARY JIMENEZ  CAPTAIN III (POLICE
DEPARTMENT)
2  3  ALBERT PARDINI  CAPTAIN III (POLICE
DEPARTMENT)
3  4  CHRISTOPHER CHONG  WIRE ROPE CABLE MAINTENANCE
MECHANIC
4  5  PATRICK GARDNER  DEPUTY CHIEF OF DEPARTMENT,(FIRE
DEPARTMENT)

   BasePay  OvertimePay  OtherPay  Benefits  TotalPay
TotalPayBenefits \
0  167411.18      0.00  400184.25      NaN  567595.43
567595.43
1  155966.02  245131.88  137811.38      NaN  538909.28
538909.28
2  212739.13  106088.18  16452.60      NaN  335279.91
335279.91
3   77916.00   56120.71  198306.90      NaN  332343.61
332343.61
4  134401.60   9737.00  182234.59      NaN  326373.19
326373.19
```

	Year	Notes	Agency	Status
0	2011	NaN	San Francisco	NaN
1	2011	NaN	San Francisco	NaN
2	2011	NaN	San Francisco	NaN
3	2011	NaN	San Francisco	NaN
4	2011	NaN	San Francisco	NaN

**** Use the .info() method to find out how many entries there are.****

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148654 entries, 0 to 148653
Data columns (total 13 columns):
Id                148654 non-null int64
EmployeeName      148654 non-null object
JobTitle          148654 non-null object
BasePay           148045 non-null float64
OvertimePay       148650 non-null float64
OtherPay          148650 non-null float64
Benefits          112491 non-null float64
TotalPay          148654 non-null float64
TotalPayBenefits  148654 non-null float64
Year              148654 non-null int64
Notes             0 non-null float64
Agency           148654 non-null object
Status            0 non-null float64
dtypes: float64(8), int64(2), object(3)
memory usage: 14.7+ MB
```

What is the average BasePay ?

66325.44884050643

**** What is the highest amount of OvertimePay in the dataset ? ****

245131.88

**** What is the job title of JOSEPH DRISCOLL ? Note: Use all caps, otherwise you may get an answer that doesn't match up (there is also a lowercase Joseph Driscoll). ****

24 CAPTAIN, FIRE SUPPRESSION
Name: JobTitle, dtype: object

**** How much does JOSEPH DRISCOLL make (including benefits)? ****

```
24      270324.91
Name: TotalPayBenefits, dtype: float64
```

**** What is the name of highest paid person (including benefits)?****

```
   Id  EmployeeName  JobTitle
\
0   1  NATHANIEL FORD  GENERAL MANAGER-METROPOLITAN TRANSIT AUTHORITY

   BasePay  OvertimePay  OtherPay  Benefits  TotalPay
TotalPayBenefits \
0  167411.18          0.0  400184.25      NaN  567595.43
567595.43

   Year  Notes  Agency  Status
0  2011   NaN  San Francisco   NaN
```

**** What is the name of lowest paid person (including benefits)? Do you notice something strange about how much he or she is paid?****

```
   Id  EmployeeName  JobTitle  BasePay
OvertimePay \
148653  148654    Joe Lopez  Counselor, Log Cabin Ranch      0.0
0.0

   OtherPay  Benefits  TotalPay  TotalPayBenefits  Year  Notes  \
148653   -618.13      0.0   -618.13      -618.13  2014   NaN

   Agency  Status
148653  San Francisco   NaN
```

**** What was the average (mean) BasePay of all employees per year? (2011-2014) ? ****

```
Year
2011    63595.956517
2012    65436.406857
2013    69630.030216
2014    66564.421924
Name: BasePay, dtype: float64
```

**** How many unique job titles are there? ****

2159

**** What are the top 5 most common jobs? ****

```
Transit Operator      7036
Special Nurse        4389
Registered Nurse     3736
Public Svc Aide-Public Works 2518
Police Officer 3     2421
Name: JobTitle, dtype: int64
```

**** How many Job Titles were represented by only one person in 2013? (e.g. Job Titles with only one occurrence in 2013?) ****

202

**** How many people have the word Chief in their job title? (This is pretty tricky) ****

477

**** Bonus: Is there a correlation between length of the Job Title string and Salary? ****

```
title_len    title_len  TotalPayBenefits
title_len    1.000000    -0.036878
TotalPayBenefits -0.036878    1.000000
```

Great Job!